

AFZAL KHAN

Bangalore, India (Open to relocate to Germany)

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Summary

Electrical harness (EWIS) design engineer with 2+ years on Boeing programs (777X/737), delivering routing, installation and manufacturing-ready documentation in CATIA V5/3DEXPERIENCE and ENOVIA. Executes DMU, tolerance/clash and separation checks with disciplined change control and ICD/E-ICD updates to sustain first-time quality. Coordinates suppliers and production to reduce design churn and protect schedules; open to Germany-based OEMs and Tier-1s.

Core Skills

- **EWIS/Design:** Harness routing & installation, connector/back-shell clocking, continuity & separation rules
- **CAD/PLM:** CATIA V5, 3DEXPERIENCE, ENOVIA, legacy V4 conversion, annotation & part-attribute hygiene
- **Verification:** DMU, tolerance/clearance, clash detection, node/network validation, ICD/E-ICD, EMC codes
- **Mfg Readiness:** Harness flattening, formboard creation, WHIMST/EMA, BOM alignment, DFM/DFMEA/DFA
- **Quality/Delivery:** First-time quality, change control, audit-ready documentation, on-time release cadence
- **Domains:** Commercial aircraft interiors (galley/lav/lighting), supplier collaboration, compliance

Experience

Cyient (Client: Boeing)

August 2023 – Present

Design Engineer

Bangalore, India

- Own EWIS routing for interiors on 777X; define harness paths, clamps & brackets, and author 2D formboards in CATIA V5.
- Run DMU/tolerance/clash and node/network checks; apply separation rules and back-shell clocking for connectivity.
- Publish E-ICD updates and EMC transfers; drive change control and audit-ready documentation across releases.
- Sustain 100% on-time delivery and 98% first-time quality; cut rework by 15% via templates and standards.

San Engineering Solutions (Client: Boeing)

February 2023 – August 2023

Design Engineer

Bangalore, India

- Convert legacy V4 models/PI drawings to CATIA V5; electrify standard parts and prepare routable assemblies.
- Route bundles and place connectors/back-shells/clamps; apply installation notes & separation codes for continuity.
- Generate EMA reports and resolve errors; maintain attribute/naming hygiene to stabilize downstream processes.
- Rebuild WHI (BCA→BDS) and replace parts; improve cycle time by ~10% and manufacturability.

Skill Lync

April 2022 – April 2023

Intern

Chennai, India

- Practice sheet-metal design and GD&T on hood/fender/backdoor; apply packaging rules to meet constraints.
- Develop engine/backdoor harness routing studies; document design assumptions, checks and decisions.
- Support reviews with organized models/drawings; build habits for clean annotations and version control.

Projects

EWIS Formboard Package Optimization

- Standardize callouts/notes across drawings reduce revision churn by 20% and improve shop clarity.
- Align WHIMST/BOM checks with annotated views; cut post-release queries and accelerate approvals.
- Template annotations to prevent drift; enable consistent manufacturing-ready documentation.

Legacy V4 →V5 Harness Migration

- Create checklist for attribute mapping, renaming and separation codes; boost conversion throughput by 25%.
- Validate routability and part integrity post-conversion; lower error rates feeding downstream tasks.
- Capture lessons learned to guide future migrations; stabilize quality across batches.

Education

Rajiv Gandhi Proudhyogiki Vishwavidyalaya

2017-2021

B.Tech, Mechanical Engineering

Bhopal, Madhya Pradesh, India

Certifications

- WES Academic Qualification Certified
- Skill Lync: Plastics, Sheet-metal & Harness Component Design of EV
- Product Lifecycle Management — Great Learning Academy
- Chancenkarte (Opportunity Card) Visa

Languages

- English – Full Professional Proficiency (C1)
- German – Intermediate Working Proficiency (B1)